Delhi Public School, R.K.Puram, New Delhi

Computer Science Practical List 2

1. Write a Python code to accept Marks of English and Maths in two separate lists E and M for 5 students, create another list as T to have the total marks as sum of corresponding marks of E and M, display the following (without using Builtin Functions/methods):
   a. Content of E
   b. Content of M
   c. Content of T
   d. Minimum marks of E, M and T
   e. Maximum marks of E, M and T
   f. Average marks of E, M and T

2. Write a Python code to perform the following:
   a. To initialize a list **`A=[10,20,30,40]`**
   b. To initialize a list **`B=[35,25,15,45]`**
   c. To create a list C to get the content from A and B as **`[10,35,20,25,30,15,40,45]`**
   d. To display content of C

3. Write a Python code with the following functions:
   a. AddValues(L) - To add integer values entered by the user in the list L
   b. DispValues(L) - To display values of L
   c. RevContent(L) - To re-arrange content of L by reversing the order of values
   d. AddEven(L) - To add all the even values of L and display the sum
   e. AddOdd(L) - To add all the odd values of L and display the sum
   f. AddZeroEnd(L) - To add all the values of L, which are ending with 0 and display the sum
   g. Show7Ten(L) - To display those values of L, which have 7 at tens place
   h. Show20_50(L) - To display those values of L, which are between 20 and 50.

   Also, call the above function in the order as a, b, c, b, d, e, f, g and h

4. Write a Python code with the following functions:
   a. AddValues(L) - To add integer values entered by the user in the list L
   b. DispValues(L) - To display values of L
   c. SwapPair(L) - To re-arrange the content of L by swapping each pair of adjacent neighbouring values
   d. SwapHalf(L) - To re-arrange content of L by swapping first half of the list with second half of the list

   Also, call the above functions in the order as a, b, c, b, d and b.

5. Write a Python code with the following functions:
   a. AddCity(C) - To add names of cities (as strings) entered by user in the list C
   b. AllUcase(C) - To change the names of cities in uppercase in the list C
   c. ShowDisp(C) - To display names of cities from list C
   d. Arrange(C) - To arrange the names of cities from list C in alphabetical order (descending).
   e. ShortNameCities(C) - To display the names of those cities from list C, which have 4 or less number of characters
   f. BigNameCities(C) - To display the names of those cities from list C, which have more than 4 characters
   g. CityNameLength(C) - To display number of alphabets present in each name of the city in list C

   Also, call the above functions in the order as a, c, b, c, d, c, e, f and g

6. Write a Python code with the following functions:
   a. AddBooks(B) - To add titles of books (as strings) entered by user in the list B
   b. AllIndia(B) - To find and display titles of all the books, which contain "INDIA" or "India" as part of its title.
   c. ShowBooks(B) - To display titles of books from list B
   d. SingleWords(B) - To display the uppercase equivalent of single word titles of books from list B.
   e. CountSingle(B) - To display the number of books from the list B which contain only a single word in its title.
   f. ThetoA(B) - To change the presence of the word “The” to “A” in the titles of the list B.

   Also, call the above functions in the order as a, c, b, d, e and f

7. Write a Python code to perform the following on a tuple **`T=(10,40,20,30,50,70)`**
   a. To display the content of T in reverse order
   b. To add and display the sum of values stored in T
   c. To find and display minimum and maximum values present in T
   d. To display sum of each adjacent pair of values
   e. To find those pair (any pair 1st-2nd, 1st-5th, 3rd-6th,...) of values from the content of T, whose sum is same as one of the values in T

8. Write a Python code to perform the following:
   a. To initialize a tuple WD containing **`(1,2,3,4,5,6,7)`**
   b. To initialize a list WDN containing **`['SUN','MON','TUE','WED','THU','FRI','SAT']`**
   c. To create a dictionary W with key-value pairs with corresponding values from WD and WDN
   d. To display content of W
   e. To re-arrange the content of dictionary in such a way that it becomes as follows: **`{1:'MON',2:'TUE',3:'WED',4:'THU,5:'FRI',6:'SAT',7:'SUN'}`**
   f. To display the content of W
   g. To copy the partial content of W in dictionaries MyDays and OfficeDays, MyDays should have content from keys 2,4 and 7 and rest from W to be the content of OfficeDays.
   h. To display the contents of MyDays and OfficeDays

9. Write a Python code to perform the following operations:
   a. To initialize a tuple TL=('RED','YELLOW','GREEN')
   b. To accepts names of 10 colors from user and store them in a list CL
   c. To display the color names from CL along with corresponding message "TRAFFIC LIGHT" and "NOT TRAFFIC LIGHT" after checking from the content of TL
   d. To initialize another tuple **`TM=('STOP','BE READY TO START/STOP','GO')`**
   e. To create a dictionary TLM by combining corresponding key-value pairs from TL and TM.
   f. To display the content of TLM

General Instructions:

i. Type the solution of all the problems using **Python** Language on **Python IDLE** or **Colab**

ii. Type the following on top of each of your program as comment lines

```
"""
Program List : <Practical Assignment List number appears here>
Program No   : <Question number in the list>
Developed By : <Your name appears here>
Date         : <Date when the program was coded appears here>
"""
```

iii. On successful execution with meaningful data, **copy** and **paste** the sample **meaningful** output at the bottom of the program as comment lines.

iv. Add all the codes in separate pages of a single Google Doc named as in the format as **`<Class>-<Section>-<L2>-<Name>`** For example **`XII-F-L2-Jai Mathir`** **(If the Name is Jai Mathir of XII-F)**

NOTE: Recommended Font, "`Courier New`", Style "`Bold`" Size "10" for all programs with single line spacing. Default indentation 2 Character only.

v. **"Turn in"** the Doc as a single document in response to this assignment submission with all programs. Once verified by the teacher, take out a hardcopy from the printer and get them signed by the respective computer teacher.

```
# ------------------------------------------------------------------#
# List-Program No    : L2-P1
# Developed By       : PUJA DEY
# Date               : 14-April-2025
# ------------------------------------------------------------------#
E=[]
for i in range(1,6,1):
  scoreE=int(input("Enter scores of English: "))
  E.append(scoreE)

M=[]
for j in range(1,6,1):
  scoreM=int(input("Enter scores of Maths:  "))
  M.append(scoreM)

T=[]
for k in range(0,5):
  T.append(E[k]+M[k])

print(E)
print(M)
print(T)

min_scoreE = E[0]
max_scoreE = E[0]
total_scoreE=0
for scoreE in E:
    if scoreE < min_scoreE:
        min_scoreE = scoreE
    if scoreE > max_scoreE:
        max_scoreE = scoreE
    total_scoreE += scoreE
average_scoreE = total_scoreE / 5

min_scoreM = M[0]
max_scoreM = M[0]
total_scoreM=0
for scoreM in M:
    if scoreM < min_scoreM:
        min_scoreM = scoreM
    if scoreM > max_scoreM:
        max_scoreM = scoreM
    total_scoreM += scoreM
```

```
average_scoreM = total_scoreM / 5

min_scoreT = T[0]
max_scoreT = T[0]
total_scoreT=0
for scoreT in T:
    if scoreT < min_scoreT:
        min_scoreT = scoreT
    if scoreT > max_scoreT:
        max_scoreT = scoreT
    total_scoreT += scoreT
average_scoreT = total_scoreT / 5

print("Min. for English: ", min_scoreE, " Min. for Maths: ", min_scoreM, " Min. for
      Total", min_scoreT)
print("Max. for English: ", max_scoreE, " Max. for Maths: ", max_scoreM, " Max. for
      Total", max_scoreT )
print("Avg score for English: ", average_scoreE, " Avg score for Maths: ",
      average_scoreM, " Avg score: ", average_scoreT,)
```

```
Enter scores of English: 7
Enter scores of English: 6
Enter scores of English: 10
Enter scores of English: 9
Enter scores of English: 8
Enter scores of Maths:  10
Enter scores of Maths:  9
Enter scores of Maths:  10
Enter scores of Maths:  9
Enter scores of Maths:  10
[7, 6, 10, 9, 8]
[10, 9, 10, 9, 10]
[17, 15, 20, 18, 18]
Min. for English:  6  Min. for Maths:  9  Min. for Total 15
Max. for English:  10  Max. for Maths:  10  Max. for Total 20
Avg score for English:  8.0  Avg score for Maths:  9.6  Avg score:  17.6
```

```
# ------------------------------------------------------------#
# List-Program No    : L2-P2
# Developed By       : PUJA DEY
# Date               : 14-April-2025
# ------------------------------------------------------------#
A=[10,20,30,40]
B=[35,25,15,45]
C=[]
for i in range(len(A)):
    C.append(A[i])
    C.append(B[i])
print(C)
```

```
[10, 35, 20, 25, 30, 15, 40, 45]
```

```
# ------------------------------------------------------------#
# List-Program No    : L2-P3
# Developed By       : PUJA DEY
# Date               : 14-April-2025
# ------------------------------------------------------------#
L = []

def AddValues(L):
    while True:
        I = int(input("Enter Integers: "))
        L.append(I)
        ch = input("Want for more integers? (Y/N): ")
        if ch.upper() == "N":
            break

def DispValues(L):
    print(L)

def RevContent(L):
    L.reverse()

def AddEven(L):
    S = 0
    for i in range(len(L)):
        if L[i] % 2 == 0:
            S = S + L[i]
    print("Sum of even values:", S)

def AddOdd(L):
    T = 0
    for j in range(len(L)):
        if L[j] % 2 != 0:
            T = T + L[j]
    print("Sum of odd values:", T)

def AddZeroEnd(L):
    U = 0
    for k in range(len(L)):
        if L[k] % 10 == 0:
            U = U + L[k]
    print("Sum of values ending with 0:", U)
```

```
def Show7Ten(L):
    for l in L:
        if (l // 10) % 10 == 7:
            print(l, end=",")
    print()

def Show20_50(L):
    for m in L:
        if 20 < m < 50:
            print(m, end=",")
    print()

AddValues(L)
DispValues(L)
RevContent(L)
DispValues(L)
AddEven(L)
AddOdd(L)
AddZeroEnd(L)
Show7Ten(L)
Show20_50(L)
```

```
Enter Integers: 20
Want for more integers? (Y/N): Y
Enter Integers: 13
Want for more integers? (Y/N): Y
Enter Integers: 50
Want for more integers? (Y/N): Y
Enter Integers: 70
Want for more integers? (Y/N): Y
Enter Integers: 17
Want for more integers? (Y/N): Y
Enter Integers: 47
Want for more integers? (Y/N): Y
Enter Integers: 82
Want for more integers? (Y/N): Y
Enter Integers: 74
Want for more integers? (Y/N): Y
Enter Integers: 1
Want for more integers? (Y/N): N
[20, 13, 50, 70, 17, 47, 82, 74, 1]
[1, 74, 82, 47, 17, 70, 50, 13, 20]
Sum of even values: 296
Sum of odd values: 78
Sum of values ending with 0: 140
74,70,
47,
```

```
# ------------------------------------------------------------#
# List-Program No    : L2-P4
# Developed By       : PUJA DEY
# Date               : 14-April-2025
# ------------------------------------------------------------#
L = []
def AddValues(L):
    while True:
        I = int(input("Enter Integer: "))
        L.append(I)
        ch = input("Want more integers? (Y/N): ")
        if ch.upper() == "N":
            break

def DispValues(L):
    print(L)
def SwapPair(L):
    for i in range(0, len(L)-1, 2):
        L[i], L[i+1] = L[i+1], L[i]
def SwapHalf(L):
  n=len(L)
  for i in range(n//2):
    L[i],L[n//2+i]=L[n//2+i],L[i]

AddValues(L)
DispValues(L)
SwapPair(L)
DispValues(L)
SwapHalf(L)
DispValues(L)
```

```
Enter Integer: 10
Want more integers? (Y/N): Y
Enter Integer: 20
Want more integers? (Y/N): Y
Enter Integer: 30
Want more integers? (Y/N): Y
Enter Integer: 40
Want more integers? (Y/N): Y
Enter Integer: 50
Want more integers? (Y/N): Y
Enter Integer: 60
Want more integers? (Y/N): Y
Enter Integer: 70
Want more integers? (Y/N): N
[10, 20, 30, 40, 50, 60, 70]
[20, 10, 40, 30, 60, 50, 70]
[30, 60, 50, 20, 10, 40, 70]
```

```
# ------------------------------------------------------------#
# List-Program No    : L2-P5
# Developed By       : PUJA DEY
# Date               : 14-April-2025
# ------------------------------------------------------------#
C = []
def AddCity(C):
    while True:
        city = input("Enter City Name: ")
        C.append(city)
        ch = input("Want more cities? (Y/N): ")
        if ch.upper() == "N":
            break
def AllUcase(C):
    for i in range(len(C)):
        C[i] = C[i].upper()
def ShowDisp(C):
    print(C)
def Arrange(C):
    C.sort(reverse=True)
def ShortNameCities(C):
    for city in C:
        if len(city) <= 4:
            print(city, end=",")
    print()
def BigNameCities(C):
    for city in C:
        if len(city) > 4:
            print(city, end=",")
    print()
def CityNameLength(C):
    for city in C:
        print(city, ":", len(city))

AddCity(C)           # a
ShowDisp(C)          # c
AllUcase(C)          # b
ShowDisp(C)          # c
Arrange(C)           # d
ShowDisp(C)          # c
ShortNameCities(C)   # e
BigNameCities(C)     # f
CityNameLength(C)    # g
```

```
Enter City Name: Delhi
Want more cities? (Y/N): Y
Enter City Name: Goa
Want more cities? (Y/N): Y
Enter City Name: Bombay
Want more cities? (Y/N): Y
Enter City Name: Paris
Want more cities? (Y/N): Y
Enter City Name: Rome
Want more cities? (Y/N): N
['Delhi', 'Goa', 'Bombay', 'Paris', 'Rome']
['DELHI', 'GOA', 'BOMBAY', 'PARIS', 'ROME']
['ROME', 'PARIS', 'GOA', 'DELHI', 'BOMBAY']
ROME,GOA,
PARIS,DELHI,BOMBAY,
ROME : 4
PARIS : 5
GOA : 3
DELHI : 5
BOMBAY : 6
```

```
# ------------------------------------------------------------#
# List-Program No    : L2-P6
# Developed By       : PUJA DEY
# Date               : 14-April-2025
# ------------------------------------------------------------#
B = []
def AddBooks(B):
    while True:
        book = input("Enter Book Title: ")
        B.append(book)
        ch = input("Want more books? (Y/N): ")
        if ch.upper() == "N":
            break
def AllIndia(B):
    for book in B:
        if "INDIA" in book.upper():
            print(book)
def ShowBooks(B):
    print(B)
def SingleWords(B):
    for book in B:
        if " " not in book:
            print(book.upper())
def CountSingle(B):
    count = 0
    for book in B:
        if " " not in book:
            count += 1
    print("Number of single word books:", count)
def ThetoA(B):
    for i in range(len(B)):
        if "The " in B[i]:
            B[i] = B[i].replace("The ", "A ", 1)
        elif "THE " in B[i]:
            B[i] = B[i].replace("THE ", "A ", 1)
    print(B)
AddBooks(B)      # a
AllIndia(B)      # b
ShowBooks(B)     # c
SingleWords(B)   # d
CountSingle(B)   # e
ThetoA(B)        # f
```

```
Enter Book Title: India Story
Want more books? (Y/N): Y
Enter Book Title: The Great Lion
Want more books? (Y/N): Y
Enter Book Title: Python
Want more books? (Y/N): Y
Enter Book Title: The Secret Garden
Want more books? (Y/N): Y
Enter Book Title: INDIA BIG
Want more books? (Y/N): Y
Enter Book Title: Cinderella
Want more books? (Y/N): Y
Enter Book Title: The Alchemist
Want more books? (Y/N): N
India Story
INDIA BIG
['India Story', 'The Great Lion', 'Python', 'The Secret Garden', 'INDIA BIG',
'Cinderella', 'The Alchemist']
PYTHON
CINDERELLA
Number of single word books: 2
['India Story', 'A Great Lion', 'Python', 'A Secret Garden', 'INDIA BIG',
'Cinderella', 'A Alchemist']
```

```
# ------------------------------------------------------------#
# List-Program No    : L2-P7
# Developed By       : PUJA DEY
# Date               : 14-April-2025
# ------------------------------------------------------------#
T = (10, 40, 20, 30, 50, 70)
print("Tuple in reverse order:", T[::-1])
total = 0
for i in T:
    total += i
print("Sum of values:", total)

min_val = T[0]
max_val = T[0]
for i in T:
    if i < min_val:
        min_val = i
    if i > max_val:
        max_val = i
print("Minimum:", min_val, "Maximum:", max_val)

print("Sum of adjacent pairs:")
for i in range(len(T)-1):
    print(T[i], "+", T[i+1], "=", T[i]+T[i+1])

print("Pairs whose sum matches a value in T:")
for i in range(len(T)):
    for j in range(i+1, len(T)):
        if T[i] + T[j] in T:
            print(T[i], "+", T[j], "=", T[i] + T[j])
```

```
Tuple in reverse order: (70, 50, 30, 20, 40, 10)
Sum of values: 220
Minimum: 10 Maximum: 70
Sum of adjacent pairs:
10 + 40 = 50
40 + 20 = 60
20 + 30 = 50
30 + 50 = 80
50 + 70 = 120
Pairs whose sum matches a value in T:
10 + 40 = 50
10 + 20 = 30
10 + 30 = 40
40 + 30 = 70
20 + 30 = 50
20 + 50 = 70
```

```
# ------------------------------------------------------------#
# List-Program No    : L2-P8
# Developed By       : PUJA DEY
# Date               : 14-April-2025
# ------------------------------------------------------------#

WD = (1, 2, 3, 4, 5, 6, 7)
WDN = ['SUN', 'MON', 'TUE', 'WED', 'THU', 'FRI', 'SAT']

W = {}
for i in range(len(WD)):
    W[WD[i]] = WDN[i]
print("Original Dictionary:", W)

W = {1:'MON', 2:'TUE', 3:'WED', 4:'THU', 5:'FRI', 6:'SAT', 7:'SUN'}
print("Re-arranged Dictionary:", W)

MyDays = {}
OfficeDays = {}
for k in W:
    if k in [2,4,7]:
        MyDays[k] = W[k]
    else:
        OfficeDays[k] = W[k]

print("MyDays:", MyDays)
print("OfficeDays:", OfficeDays)
```

```
Original Dictionary: {1: 'SUN', 2: 'MON', 3: 'TUE', 4: 'WED', 5: 'THU', 6: 'FRI', 7:
'SAT'}
Re-arranged Dictionary: {1: 'MON', 2: 'TUE', 3: 'WED', 4: 'THU', 5: 'FRI', 6: 'SAT',
7: 'SUN'}
MyDays: {2: 'TUE', 4: 'THU', 7: 'SUN'}
OfficeDays: {1: 'MON', 3: 'WED', 5: 'FRI', 6: 'SAT'}
```

```
# ------------------------------------------------------------#
# List-Program No    : L2-P9
# Developed By       : PUJA DEY
# Date               : 14-April-2025
# ------------------------------------------------------------#
CL=[]
TL=('RED','YELLOW','GREEN')
for i in range(1,11):
  C=input("Enter the name of colour: ")
  CL.append(C)
print(CL)

for i in range(len(CL)):
  if CL[i] in TL:
    print(CL[i], ":A Traffic Light")
  else:
    print(CL[i], ":Not A Traffic Light")

TLM={}
TM=('STOP','BE READY TO START/STOP','GO')
for i in range(0,3):
  TLM[TL[i]]=TM[i]
print(TLM)
```

```
Enter the name of colour: RED
Enter the name of colour: YELLOW
Enter the name of colour: BLUE
Enter the name of colour: GREEN
Enter the name of colour: ORANGE
Enter the name of colour: GREY
Enter the name of colour: BLACK
Enter the name of colour: PURPLE
Enter the name of colour: PINK
Enter the name of colour: DARK BLUE
['RED', 'YELLOW', 'BLUE', 'GREEN', 'ORANGE', 'GREY', 'BLACK', 'PURPLE', 'PINK', 'DARK
    BLUE']
RED :A Traffic Light
YELLOW :A Traffic Light
BLUE :Not A Traffic Light
GREEN :A Traffic Light
ORANGE :Not A Traffic Light
GREY :Not A Traffic Light
BLACK :Not A Traffic Light
PURPLE :Not A Traffic Light
PINK :Not A Traffic Light
DARK BLUE :Not A Traffic Light
{'RED': 'STOP', 'YELLOW': 'BE READY TO START/STOP', 'GREEN': 'GO'}
```